

EXP 5: COMPARATIVE
ANALYSIS OF DIFFERENT TYPES
OF PROMPTING PATTERNS
AND EXPLAIN WITH VARIOUS
TEST SCENARIOS

LOGESHWARI N

212225040206

SLOT = T1-S9

Scenario 1: Smart Traffic Management

Basic Prompt 1

Prompt: What is smart traffic management?

Response: Smart traffic management uses AI, sensors, and cameras to monitor road traffic and control signals automatically to reduce congestion and improve traffic flow.

Basic Prompt 2

Prompt: How does AI help reduce traffic jams?

Response: AI analyzes real-time vehicle data and adjusts signal timing based on traffic density. This helps vehicles move smoothly and reduces waiting time.

Advanced Prompt

Zero Shot Prompt 1

Prompt: Suggest a technological solution to reduce heavy traffic in cities.

Response: A smart traffic management system using AI and sensors can monitor traffic flow and automatically adjust traffic signals to reduce congestion.

Zero Shot Prompt 2

Prompt: How can technology help improve road traffic efficiency?

Response: Technology such as AI traffic monitoring, GPS navigation, and smart signals can analyze traffic conditions and guide vehicles through less crowded routes.

Few Shot Prompt 1

Prompt:

Example:

Problem: Water shortage in farms

Solution: Smart irrigation system

Problem: Heavy traffic in cities

Solution: ?

Response: Smart traffic management system using AI and real-time monitoring.

Few Shot Prompt 2

Prompt:

Example:

Scenario: Online shopping recommendations

Technology: AI

Scenario: Traffic control system

Technology: ?

Response: AI with traffic sensors and cameras.

Chain of Thoughts 1

Prompt: Explain step by step how AI reduces traffic congestion.

Response:

Cameras and sensors collect traffic data.

AI analyzes vehicle movement and congestion levels.

The system adjusts signal timings automatically.

Traffic flow improves and congestion decreases.

Chain of Thoughts 2

Prompt: Describe the working process of a smart traffic system.

Response:

Sensors detect vehicle density.

Data is sent to an AI system.

AI predicts traffic patterns.

Signals are adjusted to optimize vehicle movement.

Scenario 2: Smart Farming

Basic Prompt 1

Prompt: What is smart farming?

Response: Smart farming uses AI and sensors to monitor soil, weather, and crops to improve agricultural productivity.

Basic Prompt 2

Prompt: How does AI help farmers?

Response: AI helps farmers analyze soil moisture, weather conditions, and pest attacks to make better decisions about irrigation and fertilizers.

Advanced Prompt

Zero Shot Prompt 1

Prompt: Suggest a modern technology to increase crop production.

Response: AI-based smart farming systems that monitor soil and weather conditions to guide farmers.

Zero Shot Prompt 2

Prompt: How can farmers reduce water wastage using technology?

Response: Farmers can use smart irrigation systems that use sensors and AI to provide water only when crops need it.

Few Shot Prompt 1

Prompt:

Example:

Problem: High electricity usage

Solution: Smart energy monitoring

Problem: Low crop yield

Solution: ?

Response: Smart farming using AI and sensors.

Few Shot Prompt 2

Prompt:

Example:

Field: Healthcare

Technology: AI diagnosis

Field: Agriculture

Technology: ?

Response: AI-based smart farming.

Chain of Thoughts 1

Prompt: Explain how AI improves farming step by step.

Response:

Sensors collect soil and weather data.

AI analyzes the collected data.

The system recommends irrigation and fertilizer usage.

Farmers follow the suggestions to improve crop growth.

Chain of Thoughts 2

Prompt: Describe how smart irrigation works.

Response:

Soil moisture sensors detect water levels.

Data is sent to an AI system.

AI determines when watering is needed.

Irrigation systems activate automatically.

Scenario 3: Online Shopping Recommendations

Basic Prompt 1

Prompt: What are product recommendations in online shopping?

Response: Product recommendations are suggestions given by online shopping platforms based on a user's browsing or purchase history.

Basic Prompt 2

Prompt: Why do shopping websites recommend products?

Response: They recommend products to help customers find items they may like and improve the shopping experience.

Advanced Prompt

Zero Shot Prompt 1

Prompt: Suggest a way for online stores to help customers find products faster.

Response: Online stores can use AI recommendation systems to suggest products based on user interests.

Zero Shot Prompt 2

Prompt: How can technology personalize the shopping experience?

Response: Technology can analyze user preferences and display personalized product suggestions.

Few Shot Prompt 1

Prompt:

Example:

Platform: Movie streaming
Feature: Movie recommendations

Platform: Online shopping
Feature: ?

Response: Product recommendations based on customer behavior.

Few Shot Prompt 2

Prompt:

Example:

Service: Music app

Technology: AI recommendation

Service: Online store

Technology: ?

Response: AI product recommendation system.

Chain of Thoughts 1

Prompt: Explain how product recommendations work step by step.

Response:

The system collects user browsing data.

AI analyzes purchase and search history.

Similar products are identified.

The platform recommends these products to the user.

Chain of Thoughts 2

Prompt: Describe the process of AI recommendations in e-commerce.

Response:

User data is collected.

AI identifies patterns in preferences.

Relevant products are selected.

Recommendations are displayed on the website.

Scenario 4: Navigation and Route Planning

Basic Prompt 1

Prompt: What is navigation technology?

Response: Navigation technology helps people find the best route to reach a destination using GPS and digital maps.

Basic Prompt 2

Prompt: How do navigation apps help drivers?

Response: Navigation apps provide directions, show traffic conditions, and suggest faster routes.

Advanced Prompt

Zero Shot Prompt 1

Prompt: Suggest a technology that helps travelers reach destinations faster.

Response: GPS navigation apps that analyze traffic and suggest optimal routes.

Zero Shot Prompt 2

Prompt: How can technology help drivers avoid traffic jams?

Response: Navigation systems can monitor traffic conditions and guide drivers through alternative routes.

Few Shot Prompt 1

Prompt:

Example:

Problem: Losing way in a new city

Solution: GPS navigation

Problem: Heavy traffic on roads

Solution: ?

Response: Real-time traffic monitoring and route optimization.

Few Shot Prompt 2

Prompt:

Example:

Technology: Smart farming

Field: Agriculture

Technology: GPS navigation

Field: ?

Response: Transportation.

Chain of Thoughts 1

Prompt: Explain how navigation apps work step by step.

Response:

GPS identifies the user's location.

The system calculates possible routes.

Traffic data is analyzed.

The best route is suggested.

Chain of Thoughts 2

Prompt: Describe the process of route optimization.

Response:

Map data and traffic conditions are collected.

AI analyzes road congestion.

Alternative routes are evaluated.

The fastest route is recommended.

Scenario 5: Healthcare Diagnosis

Basic Prompt 1

Prompt: What is AI in healthcare diagnosis?

Response: AI in healthcare diagnosis uses computer systems to analyze medical data and help doctors detect diseases.

Basic Prompt 2

Prompt: How can AI assist doctors?

Response: AI can analyze medical images and patient data to identify diseases quickly and accurately.

Advanced Prompt

Zero Shot Prompt 1

Prompt: Suggest a technology that can help doctors detect diseases faster.

Response: AI-based medical image analysis systems.

Zero Shot Prompt 2

Prompt: How can technology improve healthcare diagnosis?

Response: Technology can analyze medical data, identify disease patterns, and assist doctors in making accurate diagnoses.

Few Shot Prompt 1

Prompt:

Example:

Field: Agriculture

Technology: Smart irrigation

Field: Healthcare

Technology: ?

Response: AI-based medical diagnosis systems.

Few Shot Prompt 2

Prompt:

Example:

Problem: Slow disease detection

Solution: AI diagnosis tools

Problem: Difficult medical analysis

Solution: ?

Response: AI-based medical data analysis.

Chain of Thoughts 1

Prompt: Explain step by step how AI helps diagnose diseases.

Response:

Medical data and images are collected.

AI analyzes the data.

Patterns related to diseases are detected.

Doctors use the results to confirm the diagnosis.

Chain of Thoughts 2

Prompt: Describe the working process of AI medical diagnosis.

Response:

Patient data is gathered.

AI processes the data using algorithms.

Possible diseases are identified.

Doctors review the results and make decisions.

Scenario	Prompt Type	Accuracy Quality		Depth of Response
Smart Traffic Management	Basic Prompt	4	3	2
Smart Traffic Management	Advanced Prompt	5	4	4
Smart Traffic Management	Zero-Shot Prompt	4	3	3
Smart Traffic Management	Few-Shot Prompt	4	4	3
Smart Traffic Management	Chain-of-Thought Prompt	5	5	5
Smart Farming	Basic Prompt	4	3	2
Smart Farming	Advanced Prompt	5	4	4
Smart Farming	Zero-Shot Prompt	4	3	3
Smart Farming	Few-Shot Prompt	4	4	3
Smart Farming	Chain-of-Thought Prompt	5	5	5
Online Shopping Recommendations	Basic Prompt	4	3	2
Online Shopping Recommendations	Advanced Prompt	5	4	4
Online Shopping Recommendations	Zero-Shot Prompt	4	3	3
Online Shopping Recommendations	Few-Shot Prompt	4	4	3
Online Shopping Recommendations	Chain-of-Thought Prompt	5	5	5
Navigation System	Basic Prompt	4	3	2
Navigation System	Advanced Prompt	5	4	4
Navigation System	Zero-Shot Prompt	4	3	3
Navigation System	Few-Shot Prompt	4	4	3
Navigation System	Chain-of-Thought Prompt	5	5	5
Healthcare Diagnosis	Basic Prompt	4	3	2
Healthcare Diagnosis	Advanced Prompt	5	4	4

Scenario	Prompt Type	Accuracy Quality		Depth of Response
Healthcare Diagnosis	Zero-Shot Prompt	4	3	3
Healthcare Diagnosis	Few-Shot Prompt	4	4	3
Healthcare Diagnosis	Chain-of-Thought Prompt	5	5	5